

BUSN 41210: Financial Analytics

Lecture 1: Introduction, Data Visualization, and Dimension Reduction

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Chicago Booth

September 8, 2026

Course Overview

1. Introduction, Data Visualization, and Dimension Reduction
2. Linear Regression Models
3. Transformations, Interactions, Categorical Variables, Multiple Linear Regressions
4. Model Selection Principles, Screening, Information Criteria, Stepwise Selection
5. Regularization in Linear Models: Lasso & Ridge
6. Classification, Multinomials, KNN, Sensitivity/Specificity
7. Clustering, Mixture models, K-means
8. Trees, Random Forests, Boosted Trees, Variable Importance
9. Machine Learning in Finance: Advanced Applications

Jupyter Hub

- ▶ We use Jupyter Hub for this class:
`https://jupyter-class.chicagobooth.edu/`
- ▶ Use your CNet ID and password to log on (provided that you have a Booth ID)
- ▶ Each of you has a folder. You can use shared data in this folder:
`/classes/41210_MiF_fall2026/` and run codes on the Booth server.
- ▶ You do not even need have Python installed on your personal computer.

Outline

1. Introduction

- ▶ What is Big Data?
- ▶ What is Machine Learning?
- ▶ Can Machines Learn Finance?

2. Data Visualization

3. Dimension Reduction

Data: The New...

Salesforce Q1 Earnings: CEO Benioff Calls AI Models Commodities, Data 'The New Gold'

Data isn't 'the new oil' - it's way more valuable than that

HEALTH AND HEALTHCARE SYSTEMS

Data is the new gold. This is how it can benefit everyone

Jul 29, 2020

FORBES > INNOVATION

How To Make Use Of The New Gold: Data



Sandro Shubladze Forbes Councils Member
Forbes Technology Council
COUNCIL POST | Membership (Fee-Based)



Data as the new currency

OPINION COMMENTARY [Follow](#)

Data Is the New Currency

Antitrust law fails to account for how companies exploit user information to dominate markets.

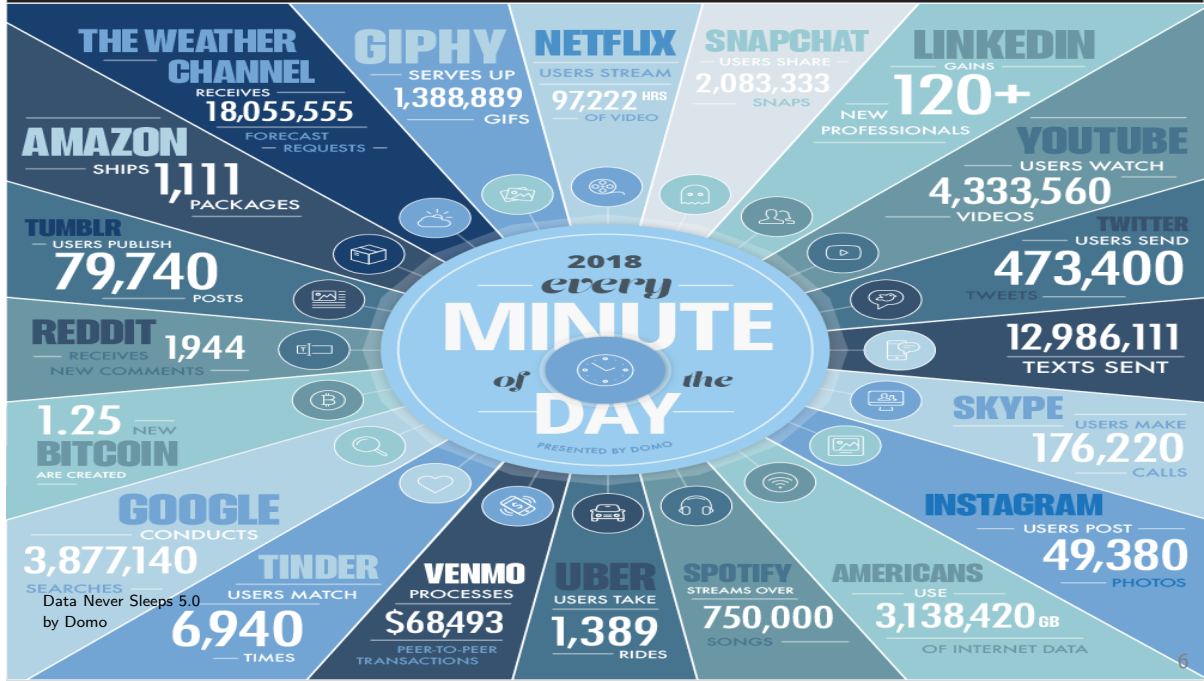
INDUSTRIES IN DEPTH



Is data the new currency?

Aug 14, 2015

WSJ



Machine Learning Rocks!

Machine Learning Successes

Machine Learning Rocks!

Machine Learning Successes

1997 Deep Blue beats Kasparov



Machine Learning Rocks!

Machine Learning Successes

1997 Deep Blue beats Kasparov

2009 Google self-driving car



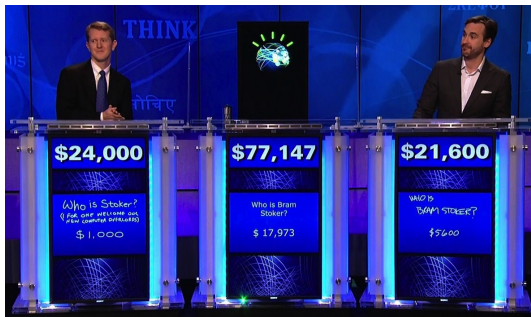
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Machine Learning Successes

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Machine Learning Rocks!

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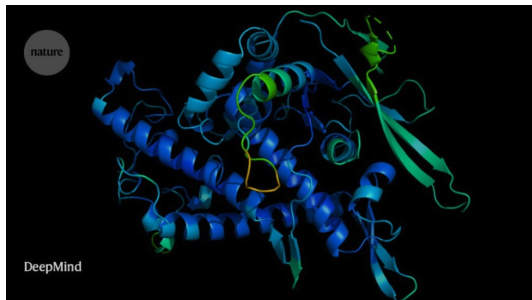
- 1997 Deep Blue beats Kasparov
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Machine Learning Rocks!

Machine Learning Successes

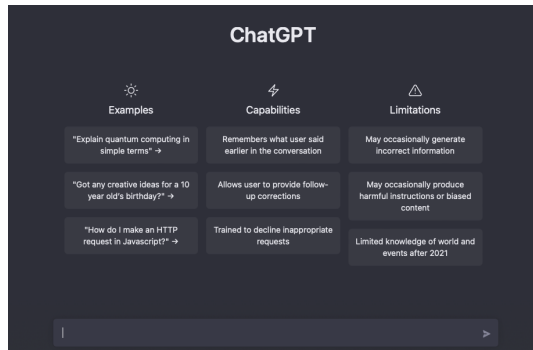
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- 2021 AlphaFold predicts protein structure



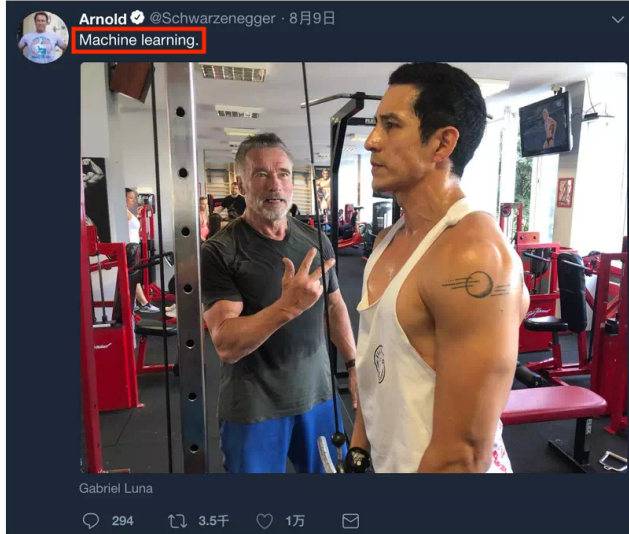
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- 2016 AlphaGo beats Lee Sedol
- 2021 AlphaFold predicts protein structure
- 2022 OpenAI's chatbot "ChatGPT"



What is “Machine Learning”?



A Paradigm Shift

There is a paradigm shift from machine *programming* to machine *learning*.

- ▶ In conventional programming, tell computer what to do, breaking big problems into many small, precisely defined tasks
- ▶ Learn (estimate) from observational data, instead of requiring pre-specified logic, for decision making and problem solving

Is this a valid email address? dacheng.xiu@chicagobooth.edu

Conventional Programming

IF upper/lowercase letters or digits

AND “@” followed by a valid domain

AND NOT special characters “!#\$...”

AND ...

THEN valid

ELSE invalid

Is this a valid email address? dacheng.xiu@chicagobooth.edu

Conventional Programming

IF upper/lowercase letters or digits
AND "@" followed by a valid domain
AND NOT special characters "!#\$%..."
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Machine Learning

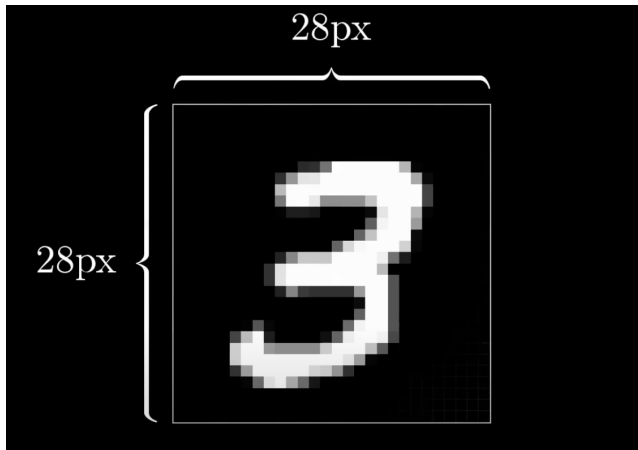
1. (Big) data on valid/invalid email addresses

Y/N	Address
0	jaime@lannister
1	hound@clegane.com
0	john.snow@GOT.edu
...	...

2. Stats Model:
$$Y/N = b_0 + b_1(@) + b_2(!\#\$\&\%?^*) \dots$$
3. Estimated probability of valid email

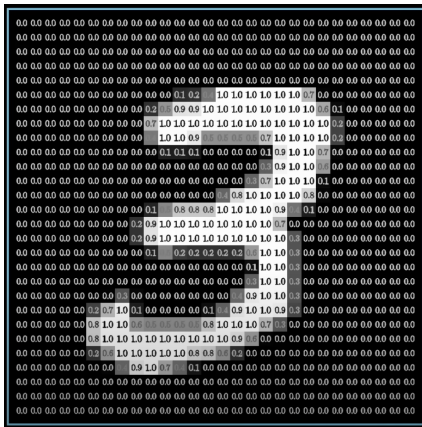
What Number is This?

Now let's try a harder example



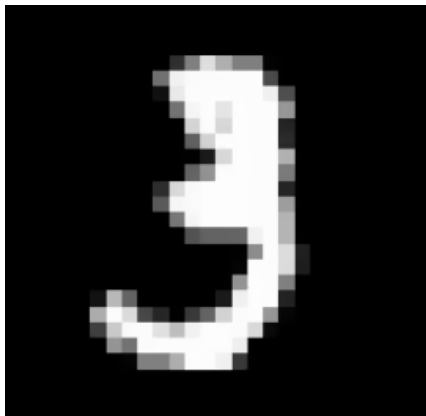
What Number is This?

- ▶ How would you even approach a problem like this, in either scenario?
- ▶ One approach could be to turn each pixel into values from 0 to 1 depending on the grayscale brightness of that pixel.



What Number is This?

- Even still, how do you code up an algorithm (a step-by-step procedure) to tell you that the previous three is the same as this three?



What Number is This?

► Or this three?



What Number is This?

- You need a flexible approach that can learn patterns of what makes a three a “three”, and that’s where machine learning shines.

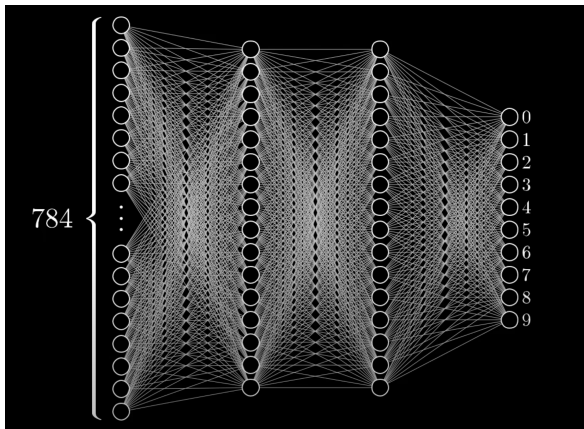


Figure: A Neural Network with 2 hidden layers, each with 16 neurons, and 10 labels

What is Machine Learning?

Arthur Samuel coined the term **Machine Learning** in his 1959 paper: “Some studies in machine learning using the game of checkers.”

Combine expertise from statistics on how to extract information from data with computational ideas that enable efficient implementation on large data sets.

Machine learning is to **big data** as human learning is to life experience.

Machine learning mimics human learning.

When a person learns to catch a ball, they do so without first being taught complicated motion equations. Instead, they practice and learn through repeated trial and error.

Machine Learning Paradigms

- ▶ Supervised learning (“predictive analytics”): a class of systems and algorithms that determine a predictive model using data points with known outcomes. Regression - The model finds outputs that are real variables; Classification - The model finds classes in which to place its inputs.
 - ▶ Predict Default; Predict Housing Prices; Predict Asset Returns; . . .
- ▶ Unsupervised learning (“exploratory data analysis”): a type of algorithm that learns patterns from unlabeled data.
 - ▶ Recommender system (Amazon - frequently bought together, products you may like, products related to this item; Netflix - Top Picks for you); Language modeling in NLP;
- ▶ Reinforcement learning: concerned with how intelligent agents make a sequence of decisions in an uncertain, potentially complex environment. In reinforcement learning, an intelligent agent faces a game-like situation. The agent employs trial and error to come up with a solution to the problem. It gets either rewards or penalties for the actions it performs. Its goal is to maximize the total reward.
 - ▶ Alpha Star playing Atari, Starcraft; Alpha Go; Self-Driving Cars; . . .

Watch a free documentary movie of AlphaGo on [Youtube](#) to gain some insight about reinforcement learning.

Can Machines Learn Finance?

Machines seem capable of anything...

- ▶ Speech recognition, translation, driving, chess/go, ...

... but can they learn finance?

The Hype

Machines taking over hedge funds despite lack of evidence they outperform humans

CNBC, July 12, 2017

Unshackled algorithms

Machine-learning promises to shake up large swathes of finance
Economist, May 2017

The Quant Fund Robot Takeover Has Been Postponed

Bloomberg, Aug 9, 2017

Return on AI

Hedge funds embrace machine learning—up to a point
Economist, Dec 2017

The U.S. Stock Market Belongs to Bots

Bloomberg, July 15, 2017

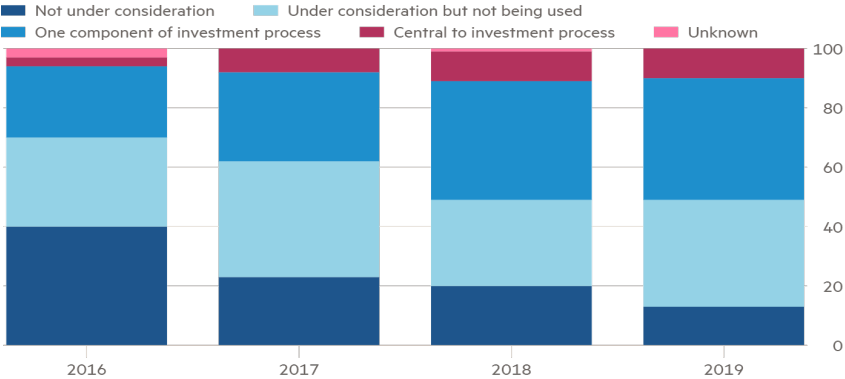


Same Reporter
3 weeks later

Machine Learning Adoption

Investment groups gradually embracing machine learning

What Morgan Stanley survey respondents said about machine learning usage



Source: Morgan Stanley
© FT

AIEQ ETF

- ▶ What is it? **Source:** <https://www.aieqetf.com/>

*AIEQ is the first actively-managed ETF to utilize **artificial intelligence** throughout the investment process. The underlying fund investments in AIEQ are based on the results of proprietary quantitative models developed by Equibot with **IBM Watson artificial intelligence**.*

AIEQ ETF

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- Performance? **Source:** Yahoo Finance



Sentient Investment

- ▶ “Will AI-Powered Hedge Funds Outsmart the Market?”

MIT Technology Review, Feb 4, 2016

*... new financial firms that advertise themselves as AI-focused may be on to something. These include **Sentient**, based in San Francisco, ...*

- ▶ “Trading, Evolved.” **Source:** <http://www.sentientim.com/about/>

*Combining **evolutionary intelligence technologies**, **deep learning algorithms** and other techniques that identify and propagate the most successful strategies, **SIM's distributed artificial intelligence system** is continuously processing and learning from **enormous stockpiles of data**, developing investment strategies in groundbreaking new ways.*

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- ▶ “AI Hedge Fund Is Said to **Liquidate** After Less Than Two Years” **Bloomberg**, Sept 6, 2018

But Why?

Finance is **Different!**

- ▶ Low signal-to-noise ratios (SNR) in sharp contrast with CS

Domain Knowledge (thanks to Gene Fama):

- ▶ **Market Efficiency:** Returns must be dominated by news in well-functioning markets
- ▶ Low SNRs are not a coincidence ...market efficiency reinforces it!

Need for Machines

“Big data” is not just about the size of the data!

- ▶ Historical data in finance (and economics) are typically not large in size, barely 10s of years, 100s of months. Lack of stationarity effectively shrinks the counts further.
- ▶ Finance does have lots of variables. With $n = 100$ data points, infeasible to run linear regressions with $p = 100$ variables.
- ▶ In cases like this ($n \approx p$), ML is inevitable.

Why Machine Learning for Finance?

- ▶ Lots of explanatory variables with potentially high correlations
- ▶ Functional form is unknown and likely complex (nonlinear)

“ML” is Inevitable

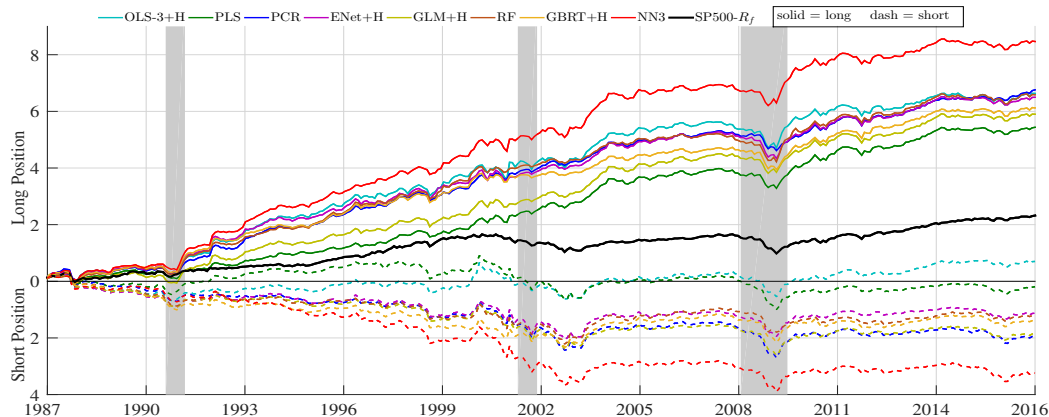
Has ML been used before? **Yes**, perhaps without knowing or emphasizing it.

“In Finance, Humans Were the First Machines,” **Chicago Booth Review, Feb 05, 2024**

- ▶ **nonlinearity**: sorting returns by characteristics
- ▶ **dimension reduction**: employing portfolios instead of individual assets
- ▶ **variable selection and factor analysis**: e.g., Fama-French factors
- ▶ **priors** (regularization): incorporating economic intuition/theory
- ▶ ...

Why not embrace modern ML techniques?

ML-based Portfolios



Source: Gu, Kelly and Xiu (2019) "Empirical Asset Pricing via Machine Learning"

Data Visualization

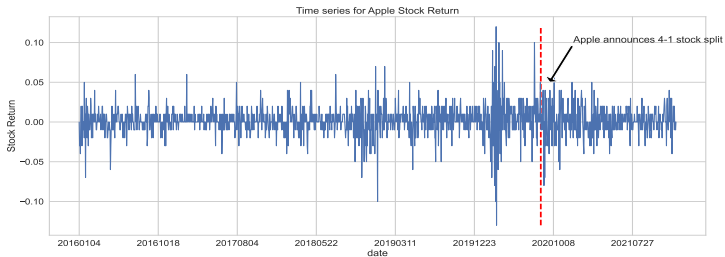
Data visualization is an essential part of exploratory data analysis (EDA)

Some guidelines:

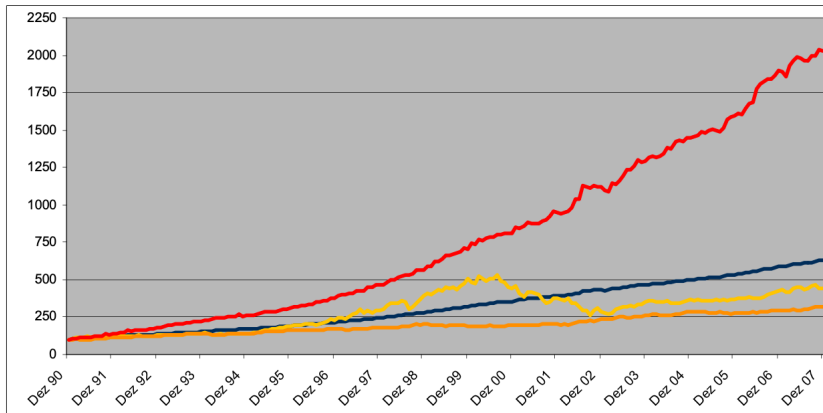
- ▶ **Statistics:** reducing dimension of your data to a few rich variables for comparison. Can be just picking two features to scatterplot, or can involve more complicated projections.
- ▶ **Design:** effective communication – with shapes, space, and color – for a given set of variable observations.
- ▶ **Language:** making it easy to move from Stats to Design.

They're all interconnected, but we'll focus on statistics.

Example: Time Series

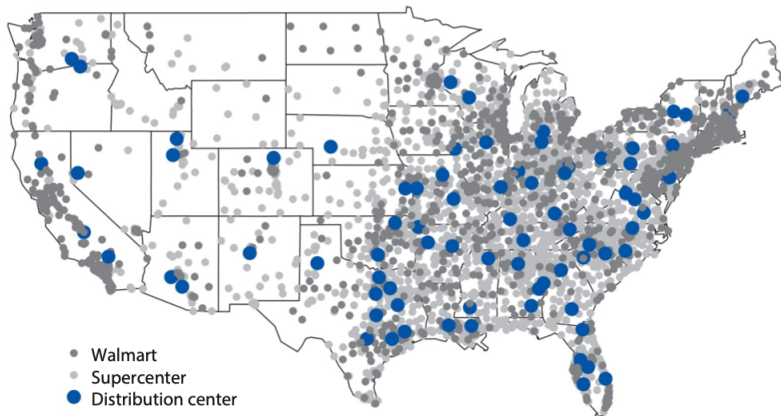


Would you like to invest in this new product (red)?



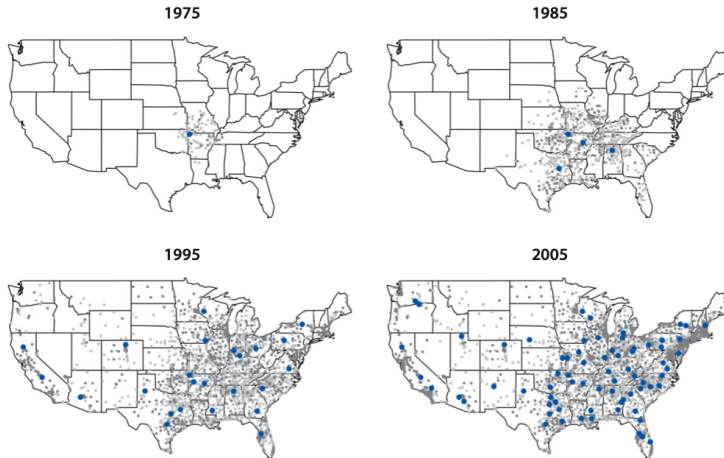
S&P 100 (yellow), Lehman Bond Index (orange), FS (blue), Simulated 3x FS (red)

Example: Spatial Data



Source: "Quantitative Social Science: An Introduction", Kosuke Imai

Example: Spatial-Temporal Data

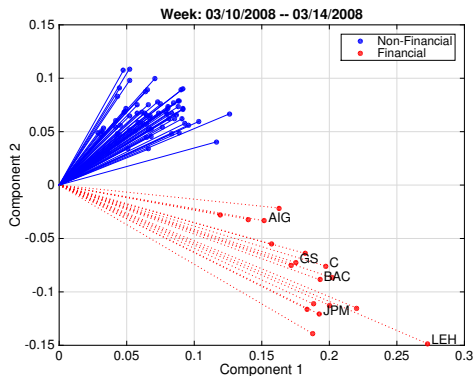
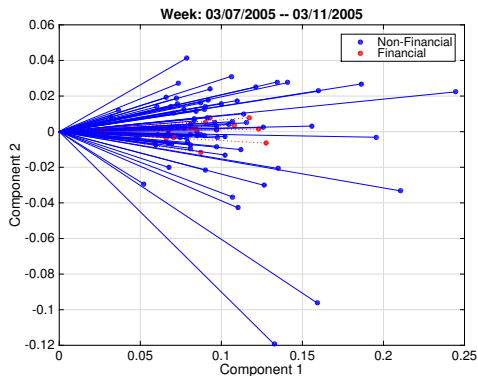


Source: "Quantitative Social Science: An Introduction", Kosuke Imai

Example: Panel Data

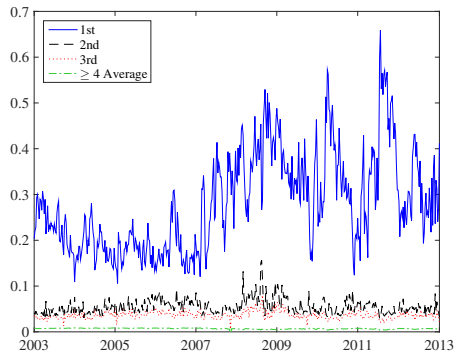
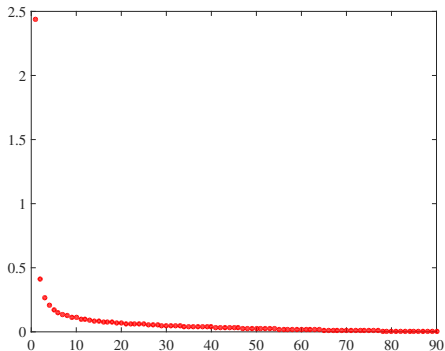
A **Biplot** is constructed by the **SVD** to obtain a low-rank approximation to the data

$$X \approx Z = UDV^T, \quad \text{where } X \text{ is } T \times n, U \text{ is } T \times 2, D \text{ is } 2 \times 2, V \text{ is } n \times 2.$$



Source: "Principal Component Analysis of High-Frequency Data", Ait-Sahalia and Xiu (2019), JASA Vol 114, No 525, 287-303.

A **scree plot** is a graph of eigenvalues against the corresponding PC number.



Source: "Principal Component Analysis of High-Frequency Data", Aït-Sahalia and Xiu (2019), JASA Vol 114, No 525, 287-303.

What “Correlated ” Means

Two variables are correlated if knowing one tells you something about the other.

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)], \quad \rho_{XY} = \frac{\text{Cov}(X, Y)}{\text{sd}(X)\text{sd}(Y)}.$$

When X is above *its* average, is Y above *its* average? Multiply the two deviations and average. Its size depends on the units, so divide by both standard deviations: correlation is covariance in units that cancel.

- ▶ ρ lies between -1 and $+1$: $+1$ a perfect line sloping up, -1 sloping down, 0 no *linear* relation.
- ▶ A correlation matrix is every pair at once — diagonal 1, symmetric, and a heat map of it is the fastest way to see structure in a wide dataset. `df.corr()`.
- ▶ $\rho = 0$ does not mean “unrelated”, only no *straight-line* relation: a perfect U-shape has $\rho \approx 0$. Look at the scatter plot as well as the number.

And it is why the next block exists: columns that are highly correlated are not carrying that many separate pieces of information.

Dimension Reduction

- ▶ $\mathbf{X}$ is a dataset with T observations of n variables.
- ▶ Each variable was previously demeaned and write

$$\mathbf{X} = (\mathbf{X}_1, \dots, \mathbf{X}_n),$$

where $\mathbf{X}_i = (X_{i1}, \dots, X_{iT})'$, for $i = 1, \dots, n$.

- ▶ Extract as much information (variance) of $\mathbf{X}$ and build another dataset $\mathbf{Z}$ with T observations but only $k \ll n$ variables:

$$\mathbf{Z} = (\mathbf{Z}_1, \dots, \mathbf{Z}_k).$$

Principal Component Analysis (PCA)

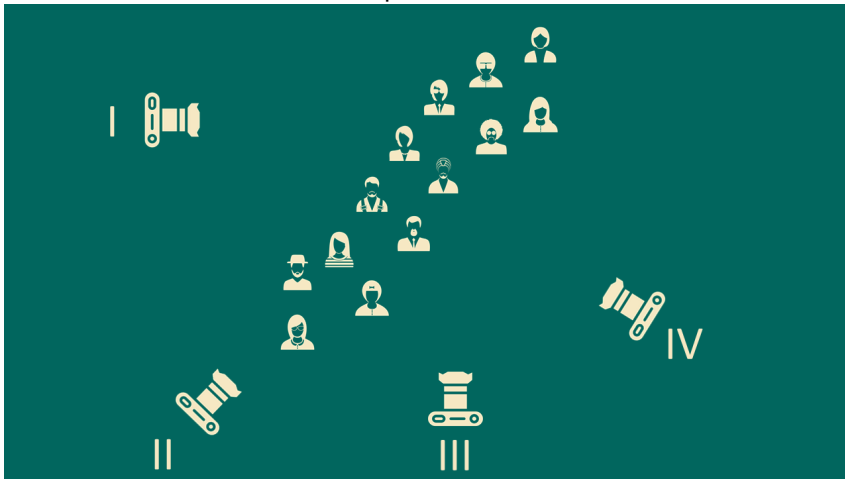
- ▶ PCA is the most common unsupervised learning approach to dimension reduction.
- ▶ PCA is a very versatile tool. It can be applied in many different situation such as:
 - ▶ Visualization of large datasets
 - ▶ Efficient use of resources (e.g.: compression)
 - ▶ Noise reduction (improving data quality)
 - ▶ Pre-treatment of the data for further machine learning algorithms
- ▶ In fact, it can be applied in any situation that requires **dimension (number of features) reduction**.
- ▶ It accomplishes such a reduction by “rotating” the dataset in a smart way.

Motivation

You want to take a picture of a group of people.



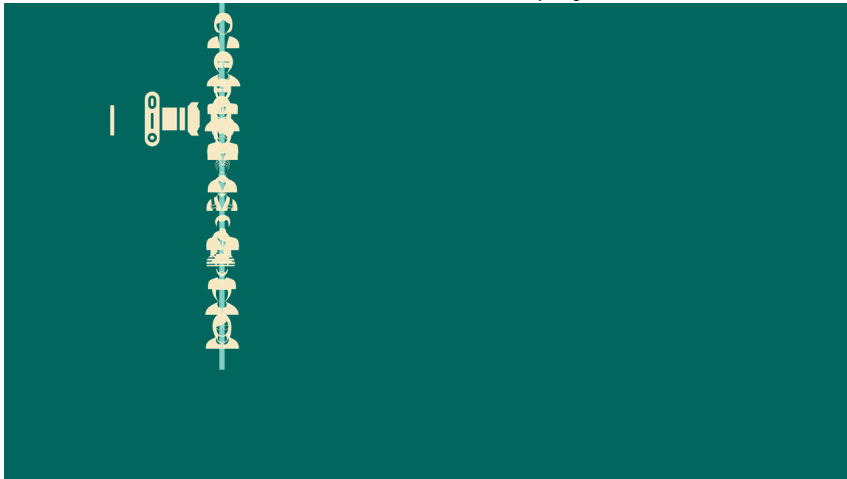
What is the best position? I, II, III, or IV?



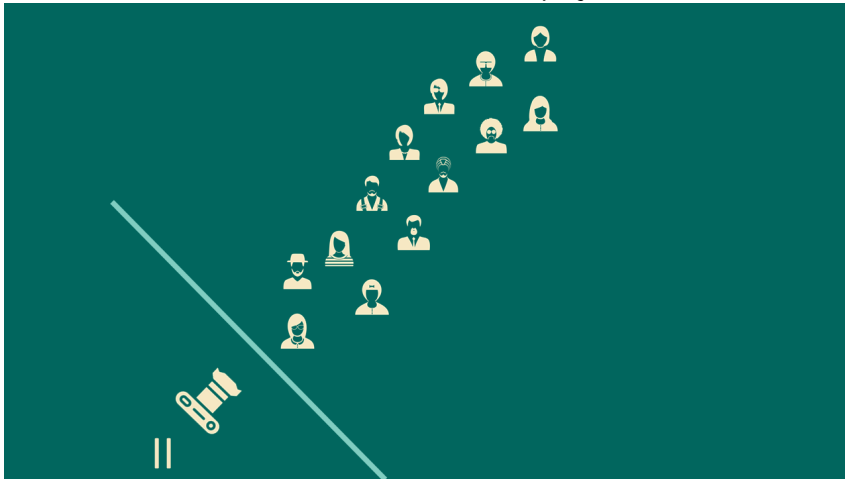
Camera 1? Let's look at the projection.



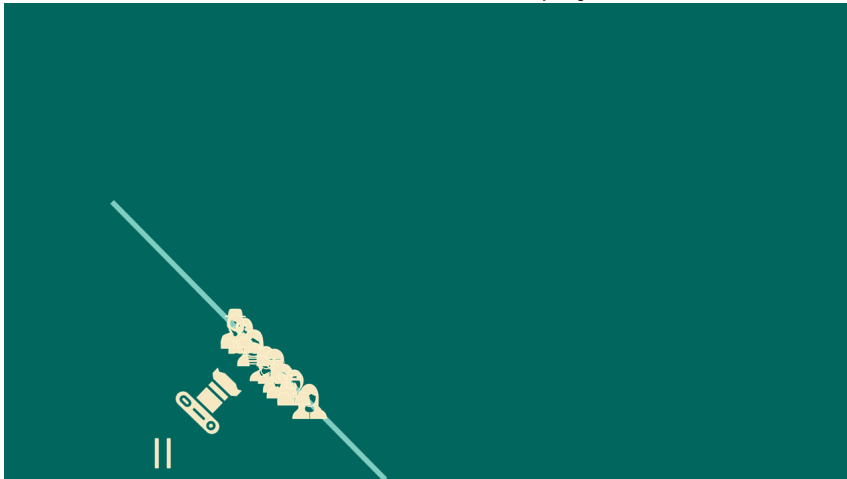
Camera I? Let's look at the projection.



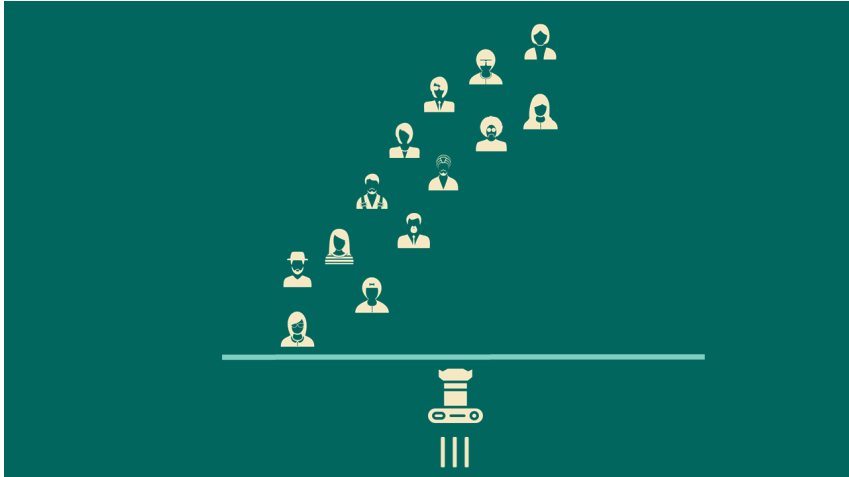
Camera II? Let's look at the projection.



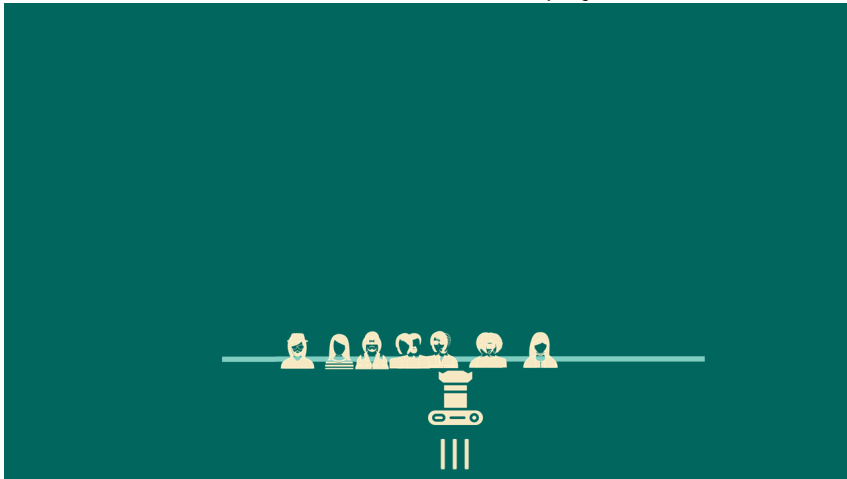
Camera II? Let's look at the projection.



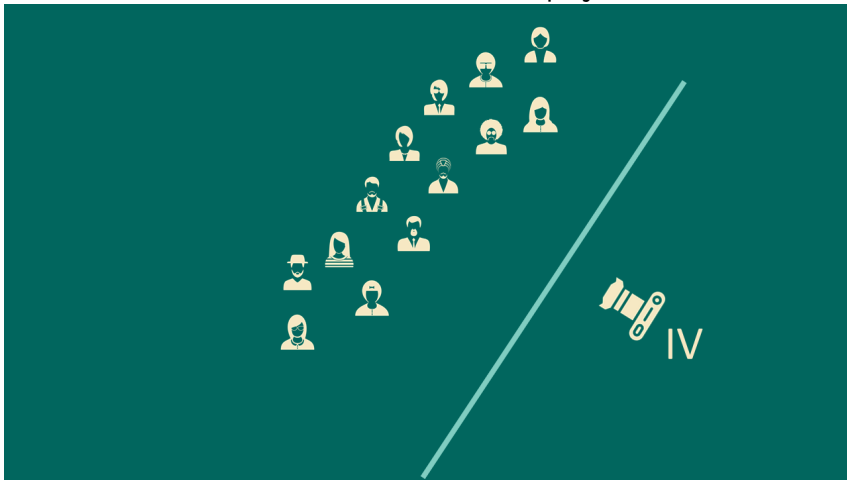
Camera III? Let's look at the projection.



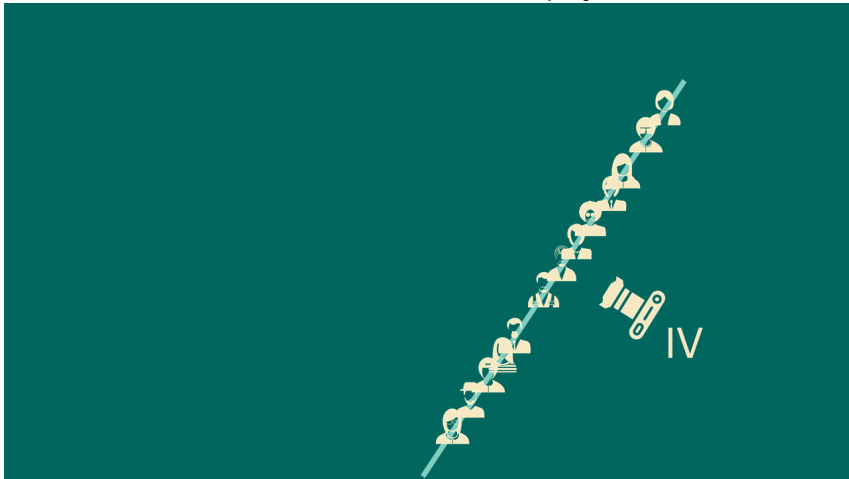
Camera III? Let's look at the projection.



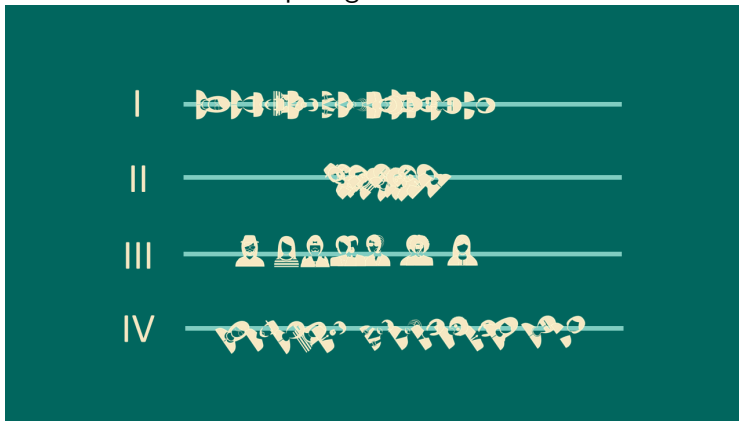
Camera IV? Let's look at the projection.



Camera IV? Let's look at the projection.



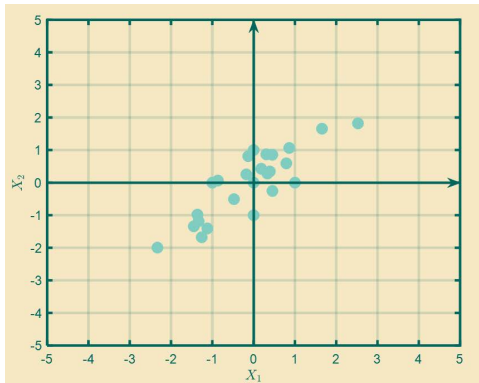
Comparing the cameras.



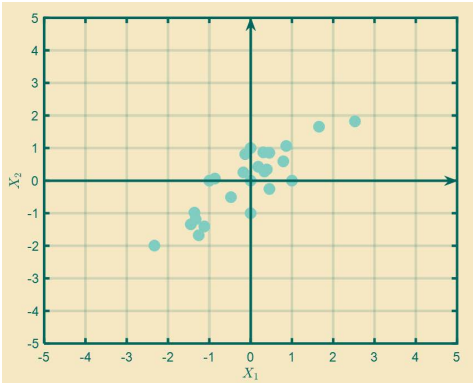
Position of camera IV seems to be the best one

How Should We Find the Best Position for the Camera?

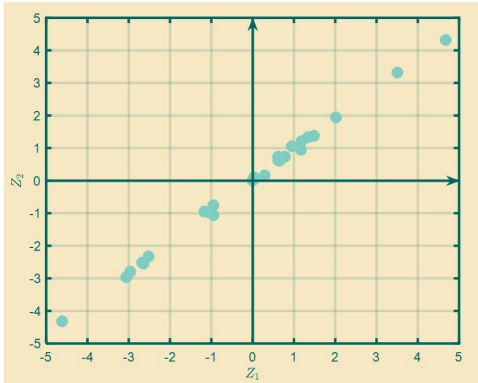
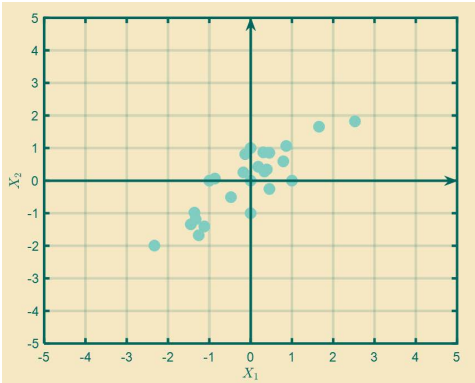
$\mathbf{X}$ is a set of points in $\mathbb{R}^2$ with sample covariance $\hat{\Sigma}$



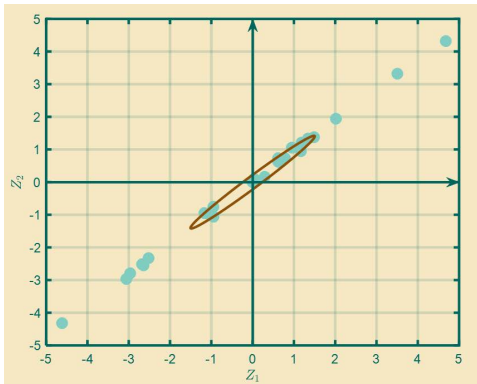
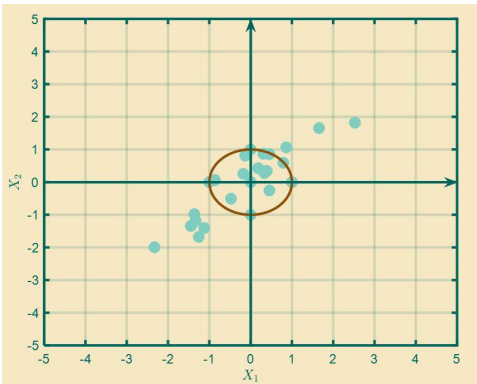
What does the transformation $\mathbf{Z} = \mathbf{X}\hat{\Sigma}$ do?



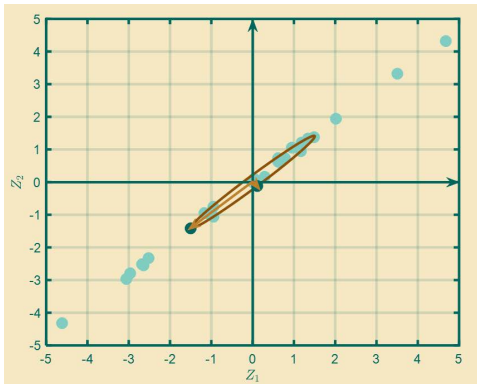
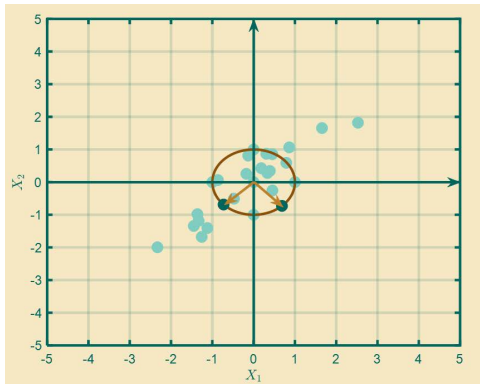
What does the transformation $\mathbf{Z} = \mathbf{X}\hat{\Sigma}$ do?



What does the transformation $\mathbf{Z} = \mathbf{X}\hat{\Sigma}$ do? Circle $\rightarrow$ ellipse.



Two special vectors: the eigenvectors. Two special scalars: the eigenvalues.



The Math of PCA

- ▶ Let $\hat{\Sigma}$ denote the $(n \times n)$ **sample covariance** matrix of $\mathbf{X}$:

$$\hat{\Sigma} = \frac{1}{T} \mathbf{X}' \mathbf{X}$$

Remember: $\mathbf{X}$ has been centered.

- ▶ Note that $\hat{\Sigma}$ is symmetric but might rank less than n if, for instance, $T < n$.
- ▶ The idea is to look for a vector $\gamma \in \mathbb{R}^n$ such that the sample variance of $\mathbf{X}\gamma$ is maximized.

- ▶ The sample variance of $\mathbf{X}\gamma$ is given by

$$\frac{1}{T}\gamma'\mathbf{X}'\mathbf{X}\gamma = \gamma'\hat{\Sigma}\gamma$$

- ▶ Without any additional constraints, we could pick a very large γ and obtain infinite variance (not useful).
- ▶ Let's look only for unitary vectors, i.e., $\|\gamma\| = 1$, or equivalently $\gamma'\gamma = 1$.

First Principal Component

- Formally, we want to solve the following constrained optimization:

$$\max_{\gamma \in \mathbb{R}^n} \gamma' \hat{\Sigma} \gamma \quad \text{s.t.} \quad \gamma' \gamma = 1.$$

- The Lagrangian is given by:

$$L(\gamma, \lambda) = \gamma' \hat{\Sigma} \gamma - \lambda(\gamma' \gamma - 1),$$

where $\lambda \in \mathbb{R}$ is the Lagrange multiplier.

First Principal Component

- ▶ Taking the derivative of the Lagrangian with respect to γ and setting equal to zero we obtain the solution:

$$\frac{\partial}{\partial \gamma_1} [\gamma_1' \hat{\Sigma} \gamma_1 - \lambda_1 (\gamma_1' \gamma_1 - 1)] = \mathbf{0}$$

$$\hat{\Sigma} \gamma_1 - \lambda_1 \gamma_1 = \mathbf{0}$$

$$\hat{\Sigma} \gamma_1 = \lambda_1 \gamma_1$$

- ▶ This is an eigenvector equation where γ_1 is an eigenvector of $\hat{\Sigma}$ and λ_1 is the associated eigenvalue.
- ▶ **Which pair of eigenvector/eigenvalue should we choose?**

- If we recognize that the quantity to be maximized is

$$\gamma_1' \underbrace{\hat{\Sigma} \gamma_1}_{=\lambda_1 \gamma_1} = \gamma_1' \lambda_1 \gamma_1 = \lambda_1 \gamma_1' \gamma_1 = \lambda_1,$$

then we should choose λ_1 to be as large as possible.

- **Solution:** pick the eigenvector associated with the largest eigenvalue of $\hat{\Sigma}$.
- We call $\mathbf{Z}_1 := \mathbf{X} \gamma_1$ the **first principal component (PC)** of $\mathbf{X}$ and note that its variance is given by λ_1 !

Second Principal Component

- ▶ Now that we have the first PC, how can we define the second?
- ▶ The second PC also comes from the maximization of $\gamma' \hat{\Sigma} \gamma$ with respect to the unitary vector γ that has to be (sample) **uncorrelated** with the first one!

- The sample covariance between the vectors $X\gamma$ and $X\gamma_1$ is

$$\begin{aligned}\frac{1}{T}(X\gamma)'X\gamma_1 &= \gamma' \frac{1}{T} \mathbf{X}' \mathbf{X} \gamma_1 \\ &= \gamma' \hat{\Sigma} \gamma_1 \\ &= \lambda_1 \gamma' \gamma_1 \quad \Leftarrow \hat{\Sigma} \gamma_1 = \lambda_1 \gamma_1\end{aligned}$$

- The restriction imposing PCs 1 and 2 to be uncorrelated translates into requiring γ and γ_1 to be orthogonal:

$$\gamma' \gamma_1 = 0$$

- Therefore, we look for the solution of

$$\max_{\gamma \in \mathbb{R}^n} \gamma' \hat{\Sigma} \gamma \quad \text{s.t.} \quad \gamma' \gamma = 1 \quad \text{and} \quad \gamma' \gamma_1 = 0$$

- In this case, the Lagrangian becomes

$$L(\gamma, \lambda, \phi) = \gamma' \hat{\Sigma} \gamma - \lambda(\gamma' \gamma - 1) - \phi \gamma' \gamma_1$$

where $\lambda, \phi \in \mathbb{R}$ are the Lagrange multipliers.

- The First Order Conditions (FOC):

$$\frac{\partial}{\partial \gamma_2} \left[\gamma_2' \hat{\Sigma} \gamma_2 - \lambda_2 (\gamma_2' \gamma_2 - 1) - \phi \gamma_2' \gamma_1 \right] = \mathbf{0}$$
$$\hat{\Sigma} \gamma_2 - \lambda_2 \gamma_2 - \phi \gamma_1 = \mathbf{0}$$

- If we left multiply the expression above by γ_1 :

$$\gamma_1' \hat{\Sigma} \gamma_2 - \lambda_2 \gamma_1' \gamma_2 - \phi \gamma_1' \gamma_1 = \mathbf{0}$$
$$0 - 0 - \phi \mathbf{1} = 0$$

- Hence, $\phi = 0$ and we obtain: $\hat{\Sigma} \gamma_2 - \lambda_2 \gamma_2 = \mathbf{0}$.

- Once again the solution is in the format of an eigenvalue equation:

$$\hat{\Sigma}\gamma_2 = \lambda_2\gamma_2$$

- The same strategy of choosing γ_2 to be the eigenvector associated with the second largest eigenvalue λ_2 yields the **second PC** of $\mathbf{X}$, namely $\mathbf{Z}_2 = \mathbf{X}\gamma_2$.
- Also the sample variance of the 2th PC is given by

$$\gamma_2'\hat{\Sigma}\gamma_2 = \lambda_2\gamma_2'\gamma_2 = \lambda_2$$

Moving Beyond the First Two Principal Components

- ▶ As you can probably guess by now, we will end up with a solution for the j -th PC of the form

$$\hat{\Sigma}\gamma_j = \lambda_j\gamma_j,$$

with γ_j orthogonal to all $\gamma_i, i = 1, 2, \dots, j - 1$.

- ▶ Process repeated until $j = n$, yielding up to n different eigenvectors $\gamma_1, \dots, \gamma_n$ of $\hat{\Sigma}$ and n eigenvalues $\lambda_1, \dots, \lambda_n$.
- ▶ The PCs are then given by

$$\mathbf{Z}_j := \mathbf{X}\gamma_j \quad , j = 1, \dots, n.$$

- ▶ Given the desired number of PCs, say $1 \leq k \leq n$, we collect all the vectors $\gamma_1, \dots, \gamma_k$ in a $(n \times k)$ matrix

$$\mathbf{\Gamma}_k := (\gamma_1, \dots, \gamma_k).$$

- ▶ The transformed dataset containing only the first k PCs is the $(T \times k)$ matrix is given by

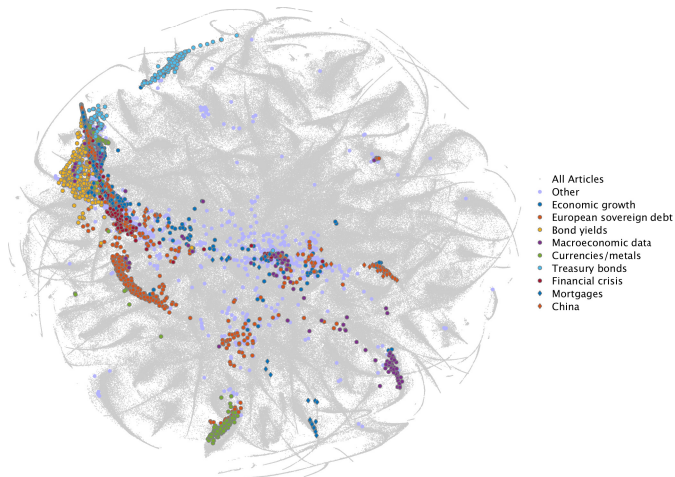
$$\mathbf{Z}_{(k)} := \mathbf{X}\mathbf{\Gamma}_k := (\mathbf{Z}_1, \dots, \mathbf{Z}_k).$$

- ▶ Also, by construction, the columns of $\mathbf{Z}_{(k)}$ (the PCs) are orthogonal random variables, and with sample variances $\lambda_1, \dots, \lambda_k$.
- ▶ Thus, the sample covariance matrix of $\mathbf{Z}_{(k)}$ is given by the $(k \times k)$ diagonal matrix

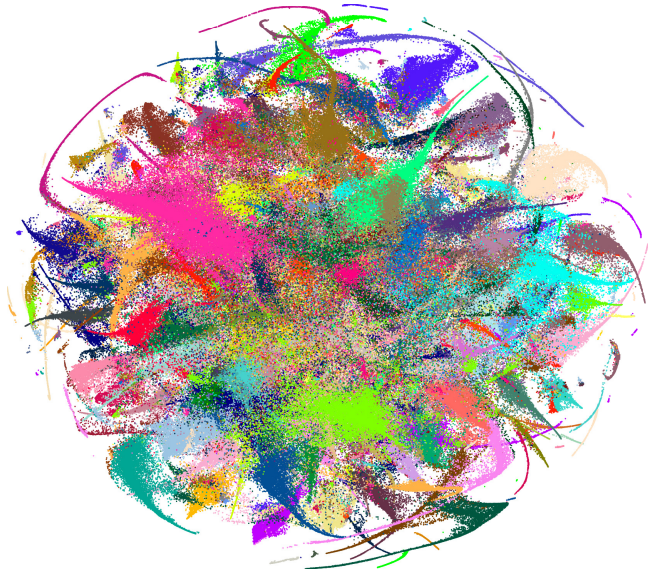
$$\mathbf{\Lambda}_k := \text{diag}(\lambda_1, \dots, \lambda_k).$$

Example: High-Dimensional Data

t-SNE visualizes **nonlinear dimension reduction** Ref: van der Maaten and Hinton (2008, JMLR).

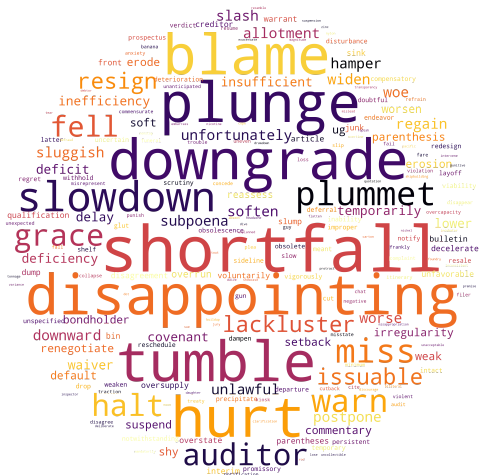
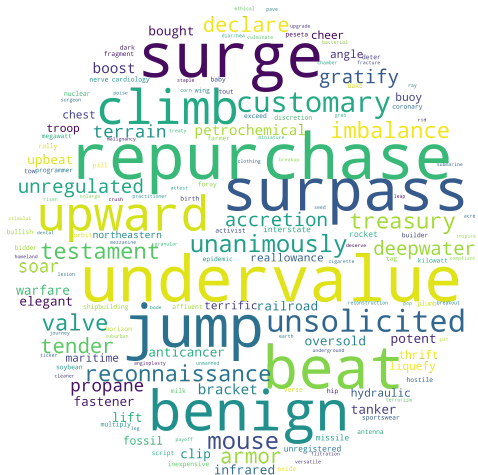


Source: "Business News and Business Cycles", Bybee, Kelly, Manela, and Xiu (2024), JF Vol 79, Issue 5, 2905-3677.



Source: "Business News and Business Cycles", Bybee, Kelly, Manela, and Xiu (2024), JF Vol 79, Issue 5, 2905-3677.

Example: Text Data

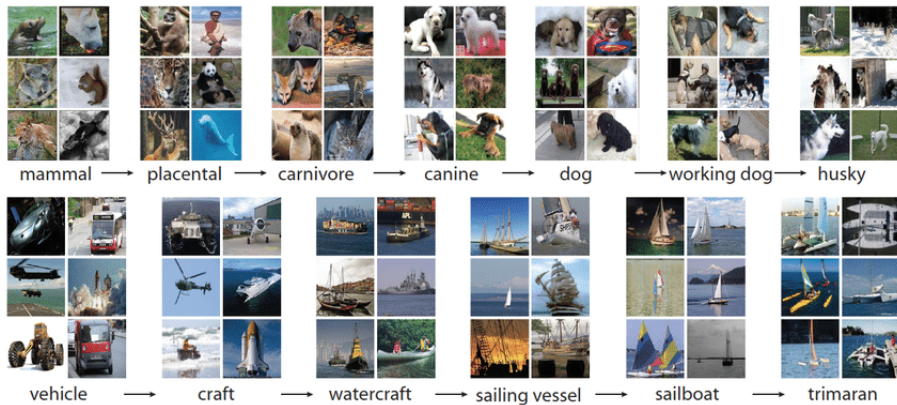


Source: "Predicting Returns with Text Data", Ke, Kelly, and Xiu (2021). 39,673 words, millions of articles.

Example: Social Media Data



Example: Image Data



ImageNet (<https://image-net.org/>): The data that transformed AI research—and possibly the world

What Has Changed, and What Has Not

These tools can now do the *mechanics* of data analysis end to end. Hand one a spreadsheet and it will load it, notice the header row is wrong, clean it, plot it, fit a model and describe the result — in about a minute, without being told the column names. That is genuinely new, and it is why this course looks different from the version taught five years ago.

Now read that list again. Not one item on it is the hard part.

- ▶ Nothing there tells you whether the answer is *right*.
- ▶ Nothing there tells you whether the question was *worth asking*.
- ▶ Nothing there tells you what would have to be *true* for the number to mean what you want it to mean.

And the same session that writes correct code will produce a confidently wrong answer in exactly the same tone. There is no tell.

So What Is Scarce Now?

When the mechanics get close to free, the value moves to the three things that are still yours:

1. **Framing.** What is the decision? What would count as an answer? A forecast, a treatment effect and a trading signal are three different questions about the same data, and they need three different setups.
2. **Validation.** How would you know if you were fooling yourself? This is most of what the next eight lectures are about, and it is the part the machine is worst at, because it requires knowing where your data came from.
3. **Judgment.** Is 0.72 good? Compared to what? A tool will tell you it is good. It does not know your benchmark.

That list is the syllabus. Nine lectures on it — and you should let the machine write the code the whole way. Lecture 9 closes the loop: the tool itself is assembled from exactly these parts.

A Live Demonstration — and What to Watch For

We hand one of these tools the macroeconomic panel from this course and ask it to predict industrial production growth. It will do something impressive. Watch for four things.

- ▶ **Does it ask what the data are?** Or does it assume? It cannot see where your file came from, and it will not usually admit that.
- ▶ **How does it split the data?** This is a monthly time series. Watch whether the future ends up in the training set. **It usually does.**
- ▶ **What does it compare the answer to?** A number with no benchmark is not a result.
- ▶ **What happens when we push back?** Tell it the split was wrong and see whether it defends the answer or folds. Both are informative.

You may use these tools on everything in this course. Lecture 0 and [AI_Coding_Guide.pdf](#) are about using them well. This slide is about why the rest of the course still exists.